

# Lesson End Project

## Automating Security Monitoring for AWS IAM Access Key Creation

**Project agenda:** To implement automated security monitoring for AWS IAM access key creation using AWS CloudTrail, Lambda, DynamoDB, and EventBridge. This ensures enhanced security by detecting and logging access key creation events and storing them in DynamoDB for auditing and tracking.

**Description:** You are a cloud engineer tasked with securing AWS resources by monitoring IAM access key creation events. The goal is to set up a system where AWS CloudTrail captures the creation of access keys, triggers an AWS Lambda function via EventBridge, and logs the event details into a DynamoDB table. This solution aims to identify potential security risks early, track malicious activity, and improve the overall security posture of the AWS account.

**Tools required:** AWS Account

**Prerequisites:** AWS account access with permissions for CloudTrail, Lambda, DynamoDB, and EventBridge, along with basic AWS and Python knowledge for setup

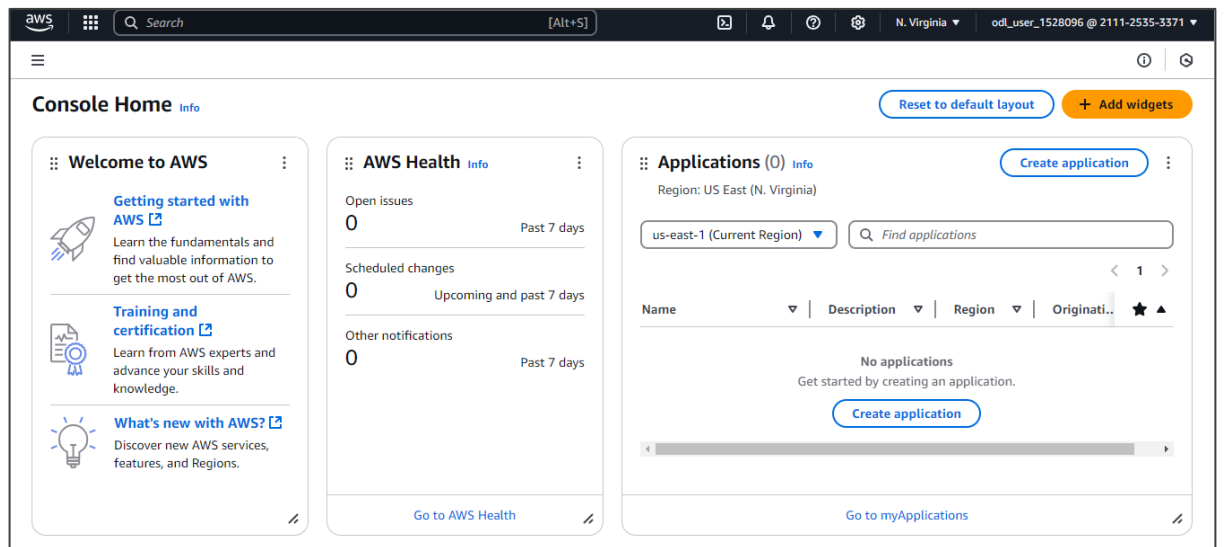
**Expected deliverables:** A security monitoring system with CloudTrail capturing key creation events, a Lambda function logging them to DynamoDB, an EventBridge trigger, and CloudWatch logs for validation

Steps to be followed:

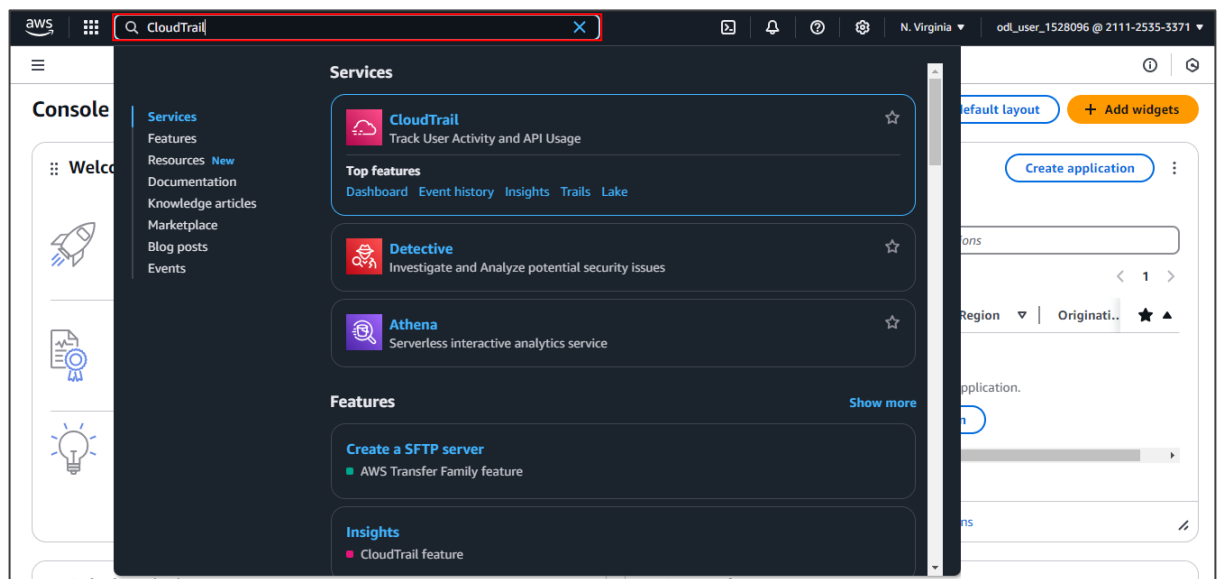
1. Create a CloudTrail to log IAM activities
2. Create and deploy a Lambda function to track access key creation
3. Create a DynamoDB table to store CloudTrail event IDs
4. Set up an EventBridge rule to trigger the Lambda function
5. Generate access and secret keys in IAM for testing
6. Monitor Lambda execution and CloudTrail logs in CloudWatch

## Step 1: Create a CloudTrail to log IAM activities

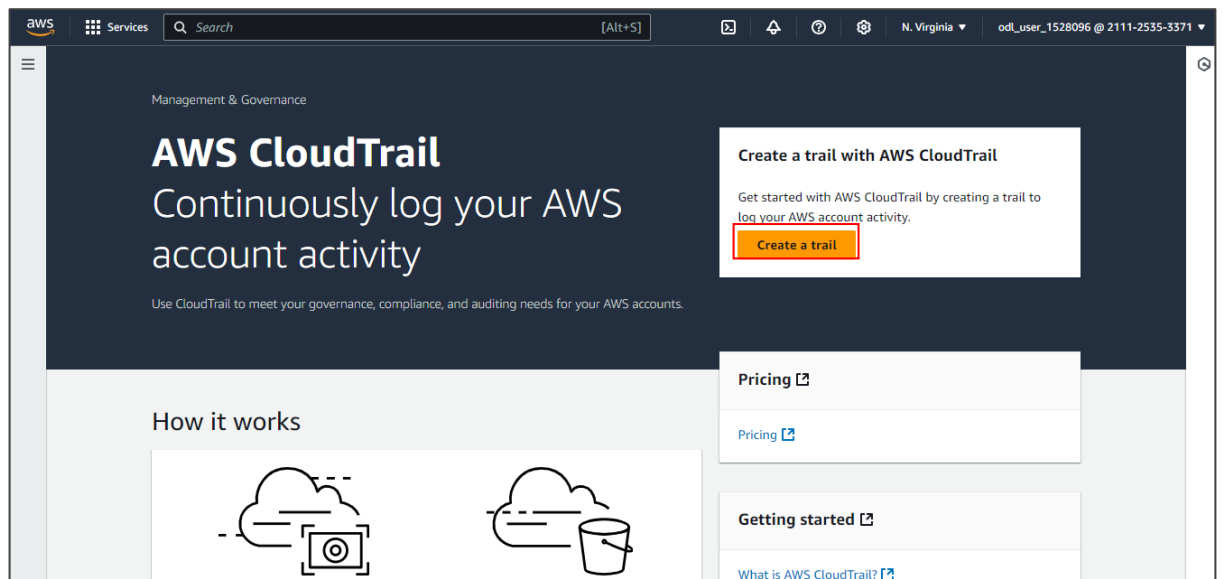
1.1 Open the Amazon console at <https://console.aws.amazon.com/>



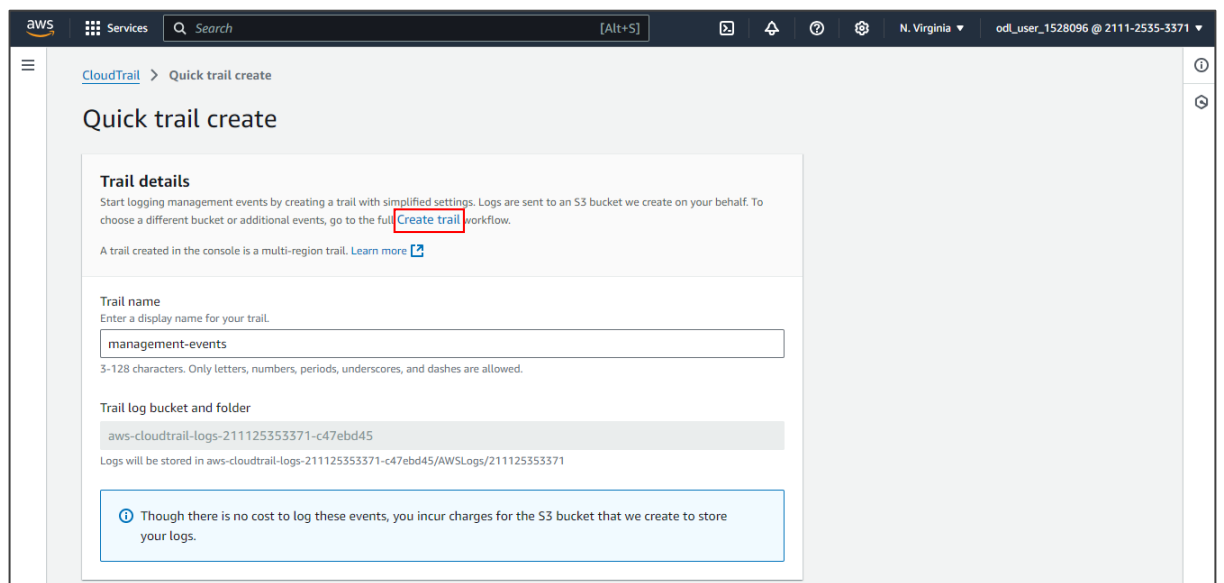
1.2 Type **CloudTrail** in the search bar and press enter to navigate to CloudTrail Dashboard



### 1.3 Click on **Create a trail**



### 1.4 Click on **Create trail** to view the complete dashboard of Cloud Trail



## 1.5 Enter the desired **Trail name** and select **Create new S3 bucket** in the Storage location section

**Choose trail attributes**

**General details**  
A trail created in the console is a multi-region trail. [Learn more](#)

**Trail name**  
Enter a display name for your trail.  
**Demo\_LEP**  
3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization  
To review accounts in your organization, open AWS Organizations. [See all accounts](#)

**Storage location** [Info](#)

☒ **Create new S3 bucket**  
Create a bucket to store logs for the trail.

☐ Use existing S3 bucket  
Choose an existing bucket to store logs for this trail.

**Trail log bucket and folder**  
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.  
**aws-cloudtrail-logs-211125353371-894ba4ac**  
Logs will be stored in `aws-cloudtrail-logs-211125353371-894ba4ac/AWSLogs/211125353371`

**Log file SSE-KMS encryption** [Info](#)  
☒ **Enabled**

**Note:** A globally unique name will be generated in the Trail log bucket and folder name, which can be changed if required.

## 1.6 Disable the checkbox **Log file SSE-KMS encryption**

☐ **Disabled**

**Additional settings**

**Log file validation** [Info](#)  
☒ **Enabled**

**SNS notification delivery** [Info](#)  
☐ **Enabled**

**CloudWatch Logs - optional**  
Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. [Learn more](#)

## 1.7 Scroll down and enable the **CloudWatch Logs**

**CloudWatch Logs - optional**  
Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. [Learn more](#)

CloudWatch Logs [Info](#)

☒ **Enabled**

Log group [Info](#)

☒ New  
☐ Existing

Log group name

aws-cloudtrail-logs-211125353371-6db9ee02

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

IAM Role [Info](#)  
AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☒ New  
☐ Existing

Role name

CloudTrailRoleForCloudWatchLogs\_{trail-name}

## 1.8 Enter the desired **Role name**

☒ New  
☐ Existing

Log group name

aws-cloudtrail-logs-211125353371-6db9ee02

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

IAM Role [Info](#)  
AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☒ New  
☐ Existing

Role name

CloudTrailRole

► Policy document

**Tags - optional** [Info](#)  
You can add one or more tags to help you manage and organize your resources, including trails.

| Key                                    | Value - optional                         |                                       |
|----------------------------------------|------------------------------------------|---------------------------------------|
| <input type="text" value="Enter key"/> | <input type="text" value="Enter value"/> | <input type="button" value="Remove"/> |

## 1.9 Scroll down and click **Next**

aws Services Search [Alt+S] N. Virginia odI\_user\_1528096 @ 2111-2535-3371

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

**IAM Role Info**  
AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☒ New  
☐ Existing

Role name  
CloudTrailRole

► Policy document

**Tags - optional** Info  
You can add one or more tags to help you manage and organize your resources, including trails.

Key Value - optional  
Enter key Enter value Remove

Add tag  
You can add 49 more tags

Cancel **Next**

## 1.10 Scroll down to **Management events** and click **Next**

aws Services Search [Alt+S] N. Virginia odI\_user\_1528096 @ 2111-2535-3371

Log the resource operations performed on or within a resource.

☐ Insights events  
Identify unusual activity, errors, or user behavior in your account.

☐ Network activity events  
Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

**Management events** Info  
Management events show information about management operations performed on resources in your AWS account.

**No additional charges** apply to log management events on this trail because this is your first copy of management events.

**API activity**  
Choose the activities you want to log.

☒ Read ☒ Write

☐ Exclude AWS KMS events  
☐ Exclude Amazon RDS Data API events

Cancel Previous **Next**

## 1.11 Scroll down in the **Review and create** page and click **Create trail**

The screenshot shows the AWS CloudTrail console in the 'Review and create' step. The left sidebar lists the steps: Step 1: Choose trail attributes (selected), Step 2: Choose log events, and Step 3: Review and create. The main content area is titled 'Review and create' and 'Step 1: Choose trail attributes'. It contains a table for 'General details' and a section for 'CloudWatch Logs'.

| General details                               |                                                                                      |                                       |
|-----------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------|
| Trail name<br>Demo_LEP                        | Trail log location<br>aws-cloudtrail-logs-211125353371-894ba4ac/AWSLogs/211125353371 | Log file validation<br>Enabled        |
| Multi-region trail<br>Yes                     |                                                                                      | SNS notification delivery<br>Disabled |
| Apply trail to my organization<br>Not enabled | Log file SSE-KMS encryption<br>Not enabled                                           |                                       |

CloudWatch Logs

|                                                        |                            |
|--------------------------------------------------------|----------------------------|
| Log group<br>aws-cloudtrail-logs-211125353371-6db9ee02 | IAM Role<br>CloudTrailRole |
|--------------------------------------------------------|----------------------------|

The screenshot shows the 'Review and create' page with the 'Step 1: Choose trail attributes' section expanded. It displays the 'Data events' section, which is currently not configured for this trail. Below it, the 'Insights events' section is also not configured. The 'Network activity events' section is also not configured. At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Create trail' (highlighted with a red border).

Data events

Data event collection is not configured for this trail

Insights events

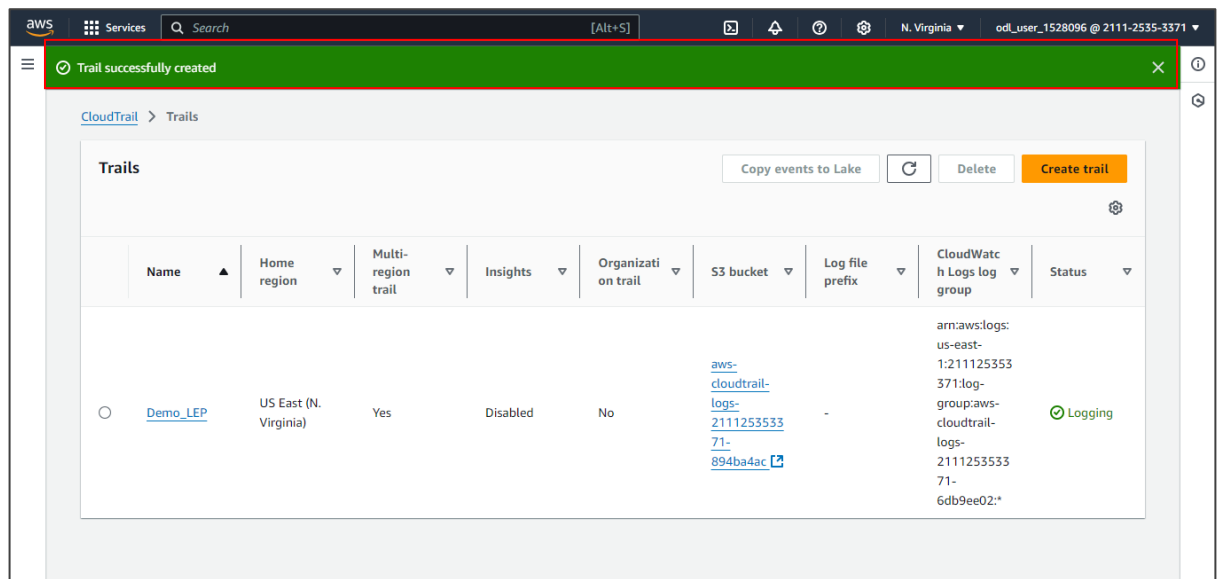
You can only enable CloudTrail Insights on trails that log management events. [Learn more](#)

Network activity events

Network activity event collection is not configured for this trail

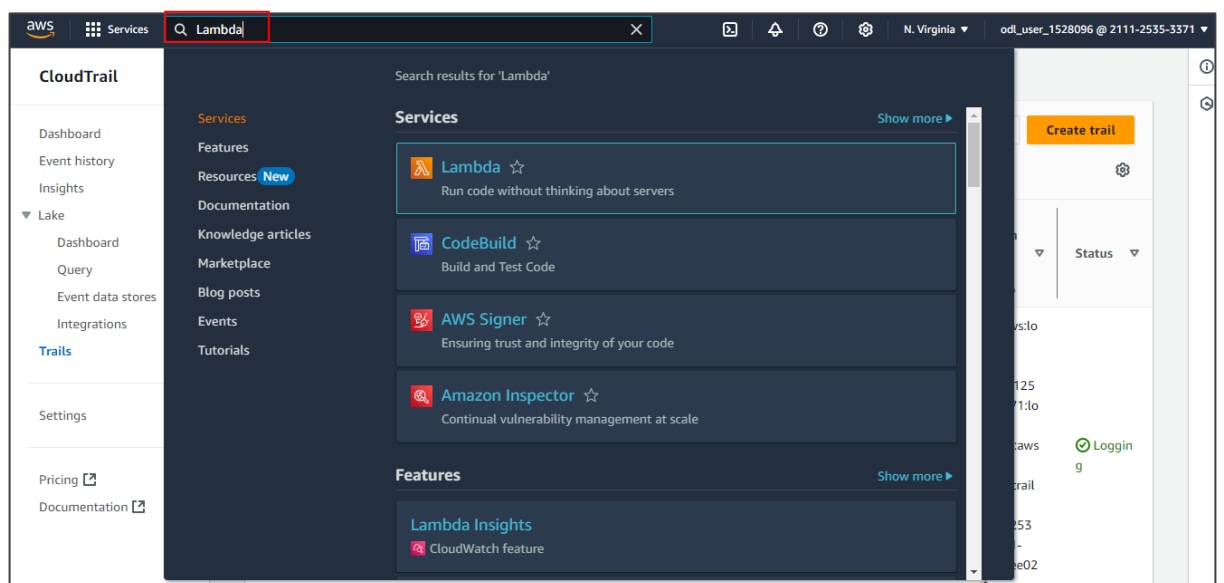
Cancel Previous **Create trail**

You will see a **Trail successfully created** message box.

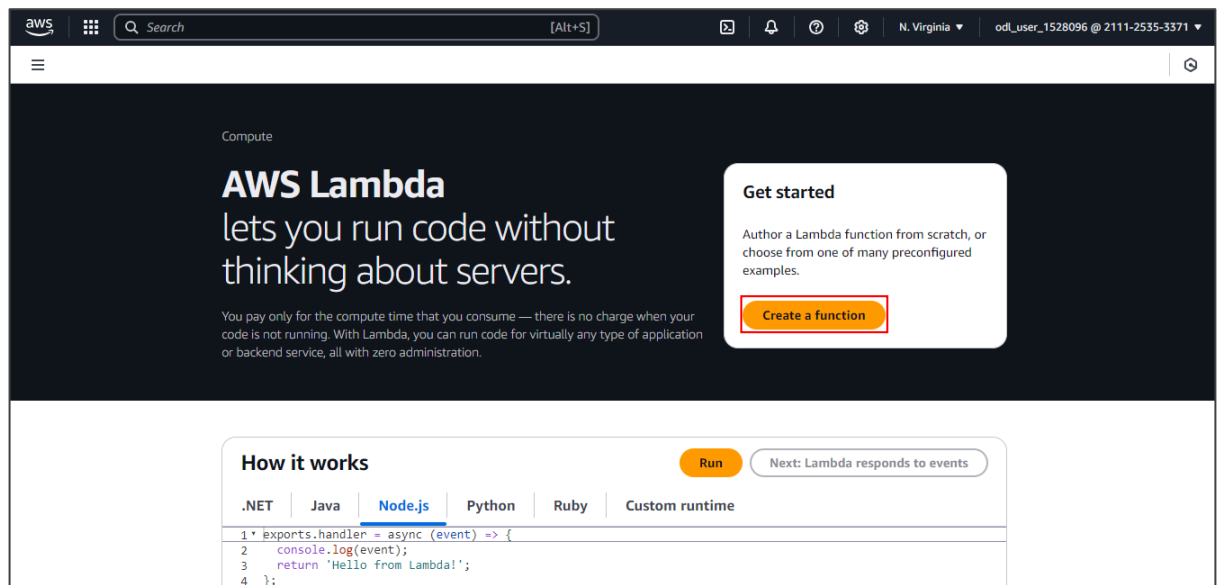


## Step 2: Create and deploy a Lambda function to track access key creation

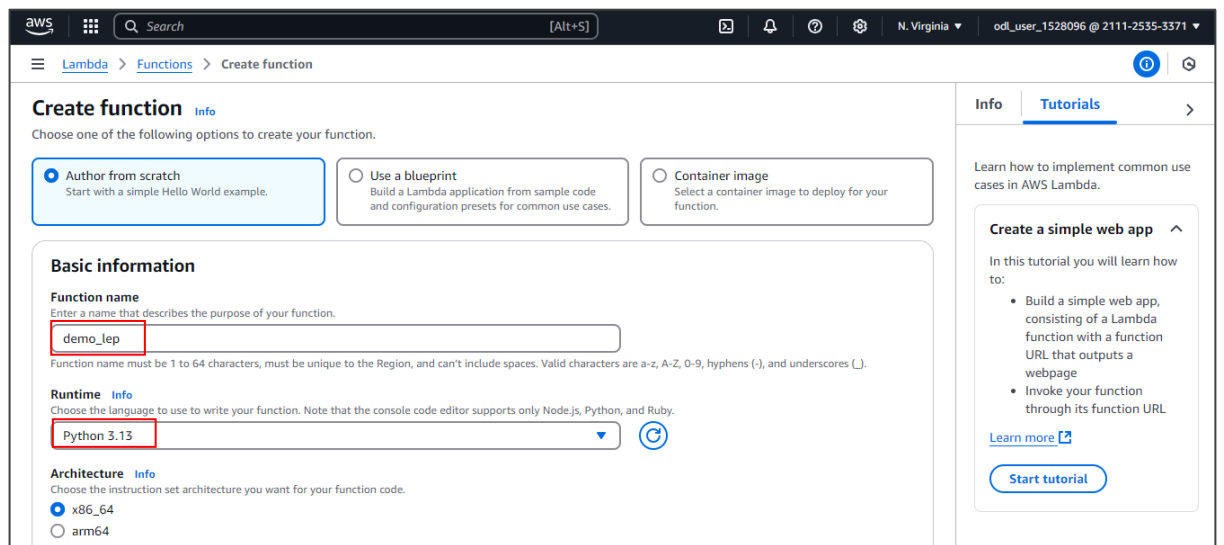
### 2.1 Type **Lambda** in the search bar and press enter to navigate to Lambda Dashboard



## 2.2 Click on **Create a function**



## 2.3 Enter a desired **Function name** and select **Python 3.13** in the Runtime section



## 2.4 Click on Create function

The screenshot shows the AWS Lambda 'Create function' page. The function name 'demo\_lep' is entered in the top field. Below it, the 'Runtime' is set to 'Python 3.13' and the 'Architecture' is set to 'x86\_64'. The 'Permissions' section is expanded, showing a 'Change default execution role' button. At the bottom right, there are 'Cancel' and 'Create function' buttons. The right sidebar contains a 'Tutorials' section with a 'Create a simple web app' tutorial link.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (\_).

**Runtime** Info  
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.13

**Architecture** Info  
Choose the instruction set architecture you want for your function code.

☒ x86\_64  
☐ arm64

**Permissions** Info  
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

► Change default execution role

► Additional Configurations  
Use additional configurations to set up code signing, function URL, tags, and Amazon VPC access for your function.

Cancel Create function

**Info** Tutorials  
Learn how to implement common use cases in AWS Lambda.

**Create a simple web app** ^  
In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

[Learn more](#) [Start tutorial](#)

You will see a message box with the following message:

The screenshot shows the AWS Lambda console for the 'demo\_lep' function. A green success message box at the top states: 'Successfully created the function demo\_lep. You can now change its code and configuration. To invoke your function with a test event, choose "Test".' The function overview section shows the function name 'demo\_lep', its layers (0), and buttons to 'Add trigger' and 'Add destination'. The right sidebar contains a 'Tutorials' section with a 'Create a simple web app' tutorial link.

Successfully created the function **demo\_lep**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

**demo\_lep** Throttle Copy ARN Actions

▼ Function overview Info  
Diagram Template

+ Add trigger + Add destination

**demo\_lep**  
Layers (0)

**Description**  
-

**Last modified**  
2 seconds ago

**Function ARN**  
arn:aws:lambda:us-east-1:211125353371:function:demo\_lep

**Function URL** Info  
-

**Info** Tutorials  
Learn how to implement common use cases in AWS Lambda.

**Create a simple web app** ^  
In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

[Learn more](#) [Start tutorial](#)

2.5 Select the **Code** tab and paste the code given below in the **lambda\_function.py** file

```
import json
from datetime import datetime, timedelta
import boto3

#Create a DynamoDB client
table_name='iamkey_storer'
region_name='us-east-1'
client_dynamo=boto3.resource('dynamodb',region_name=region_name)
dynamodb=client_dynamo.Table(table_name)

# Function to create an item in DynamoDB table
def put_item_to_dynamodb(item):
    dynamodb.put_item(Item=item)

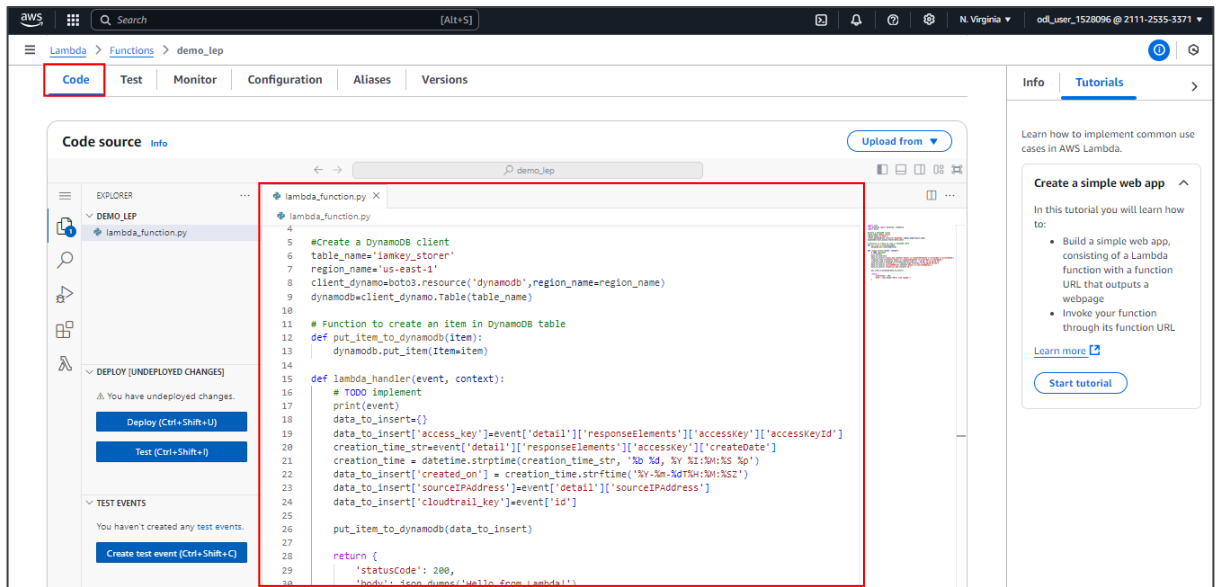
def lambda_handler(event, context):
    # TODO implement
    print(event)
    data_to_insert={}

    data_to_insert['access_key']=event['detail']['responseElements']['accessKey']['accessKeyId']

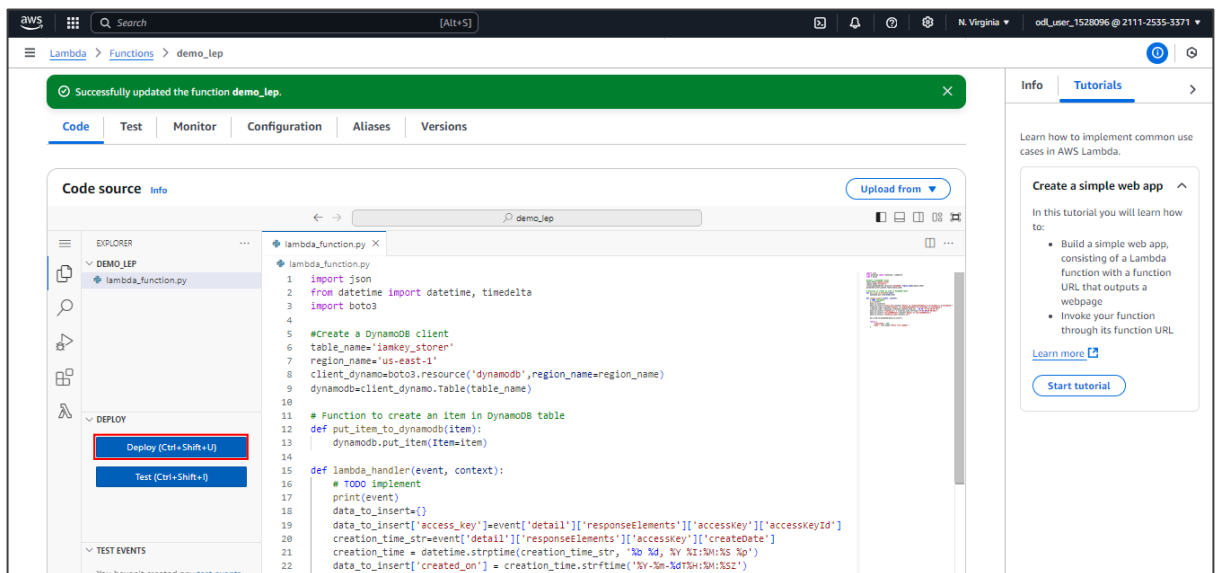
    creation_time_str=event['detail']['responseElements']['accessKey']['createDate']
    creation_time = datetime.strptime(creation_time_str, '%b %d, %Y %l:%M:%S %p')
    data_to_insert['created_on'] = creation_time.strftime('%Y-%m-%dT%H:%M:%SZ')
    data_to_insert['sourceIPAddress']=event['detail']['sourceIPAddress']
    data_to_insert['cloudtrail_key']=event['id']

    put_item_to_dynamodb(data_to_insert)

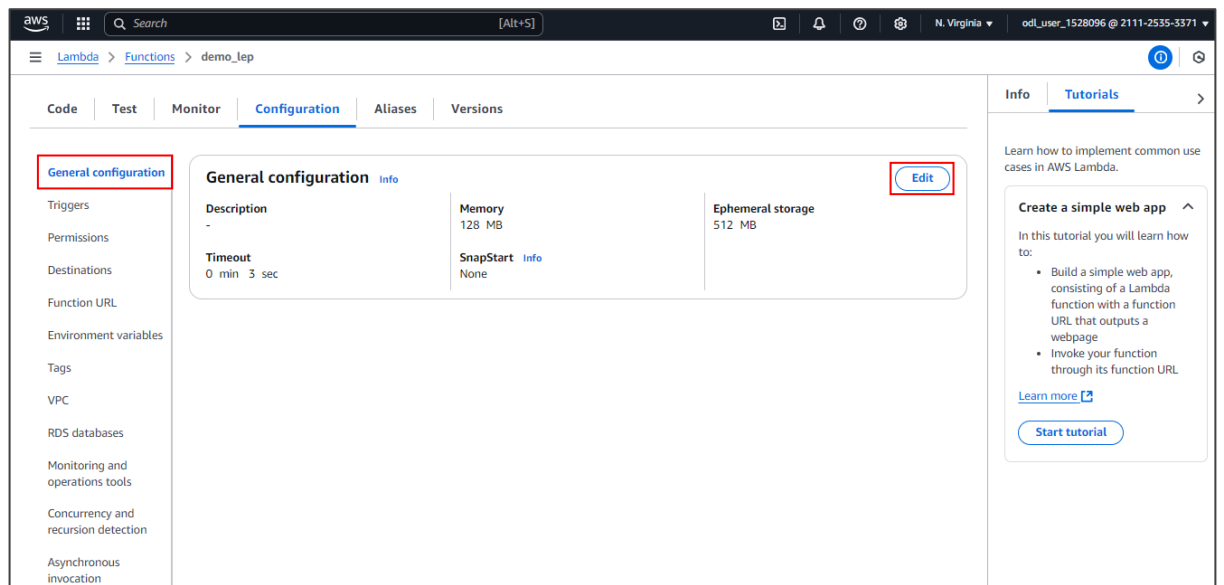
    return {
        'statusCode': 200,
        'body': json.dumps('Hello from Lambda!')
    }
```



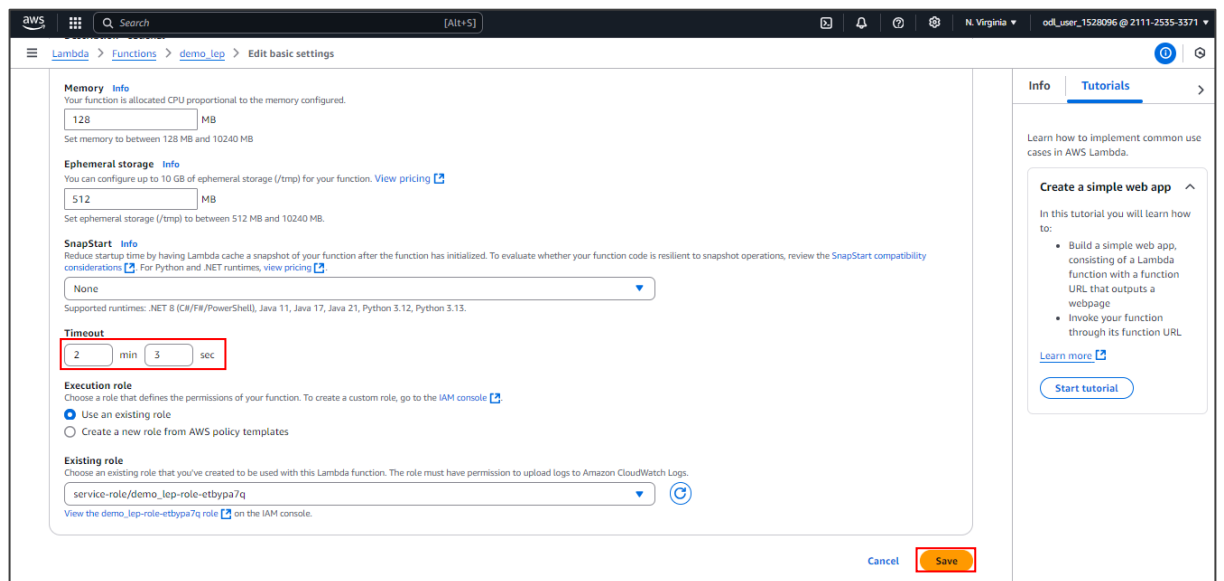
## 2.6 Click on **Deploy** to update the changes made in the `lambda_function.py` file



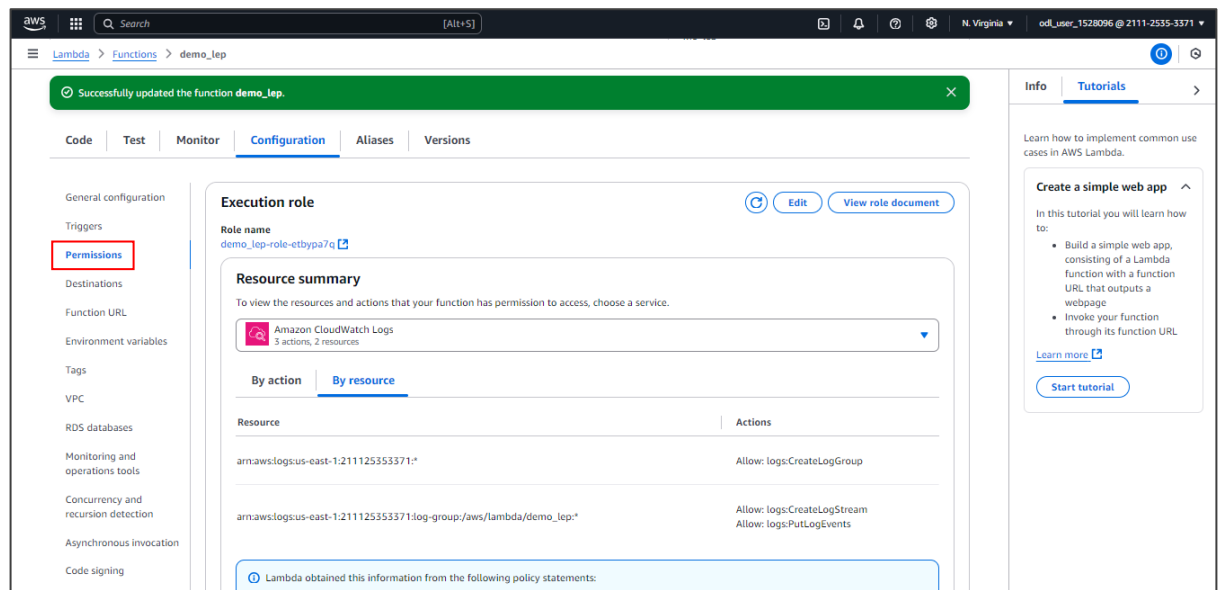
## 2.7 Select the **Configuration** tab, select **General configuration**, and click on **Edit**



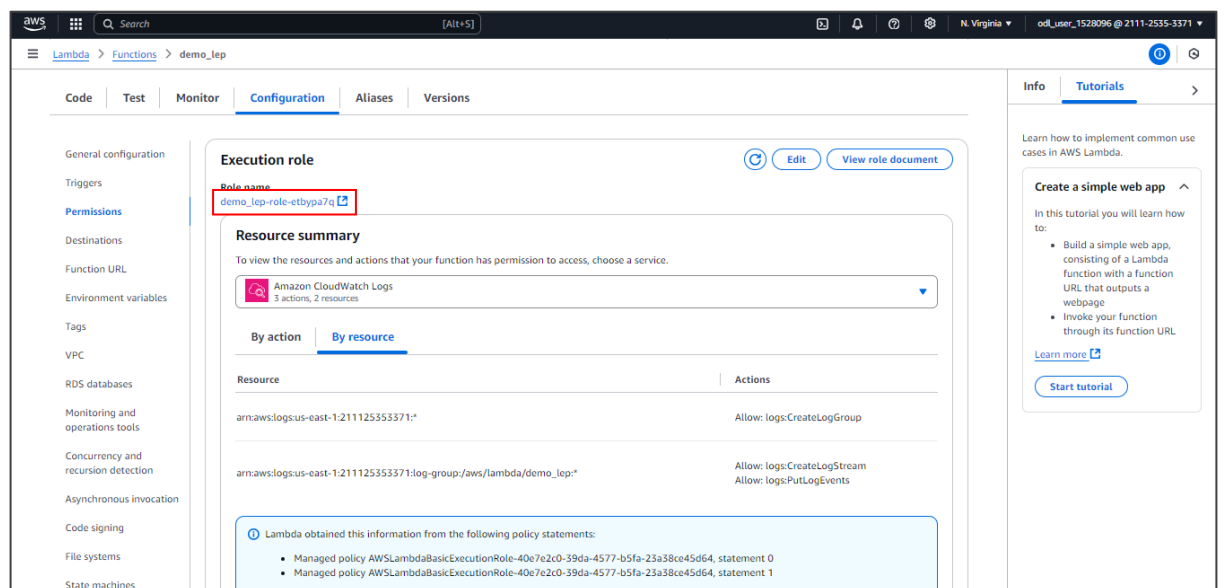
## 2.8 Change the timeout to **2 min 3 sec** and click on **Save**



## 2.9 Click on **Permissions**



## 2.10 Click on **Role name** to navigate to the IAM dashboard



## 2.11 Click on **Add permissions**

The screenshot shows the AWS IAM console interface for the role **demo\_lep-role-etbypa7q**. The left sidebar contains the navigation menu with sections like Identity and Access Management (IAM), Access management, and Access reports. The main content area shows the role's summary, including creation date (November 28, 2024, 13:09 UTC+05:30), ARN, and maximum session duration (1 hour). Below the summary, the **Permissions** tab is active, displaying a table of attached policies. The **Add permissions** button is highlighted with a red box.

**Summary**

Creation date: November 28, 2024, 13:09 (UTC+05:30)

ARN: `arn:aws:iam::211125353371:role/service-role/demo_lep-role-etbypa7q`

Maximum session duration: 1 hour

**Permissions policies (1)**

You can attach up to 10 managed policies.

| Policy name                                              | Type             | Attached entities |
|----------------------------------------------------------|------------------|-------------------|
| <code>AWSLambdaBasicExecutionRole-40e7e2c0-39d...</code> | Customer managed | 1                 |

**Permissions boundary** (not set)

**Generate policy based on CloudTrail events**

## 2.12 Select **Attach policies**

The screenshot shows the AWS IAM console interface for the role **demo\_lep-role-etbypa7q**. The left sidebar contains the navigation menu. The main content area shows the role's summary and the **Permissions** tab. The **Add permissions** button is highlighted with a red box, and the **Attach policies** option in the dropdown menu is also highlighted with a red box.

**Summary**

Creation date: November 28, 2024, 13:09 (UTC+05:30)

ARN: `arn:aws:iam::211125353371:role/service-role/demo_lep-role-etbypa7q`

Maximum session duration: 1 hour

**Permissions policies (1)**

You can attach up to 10 managed policies.

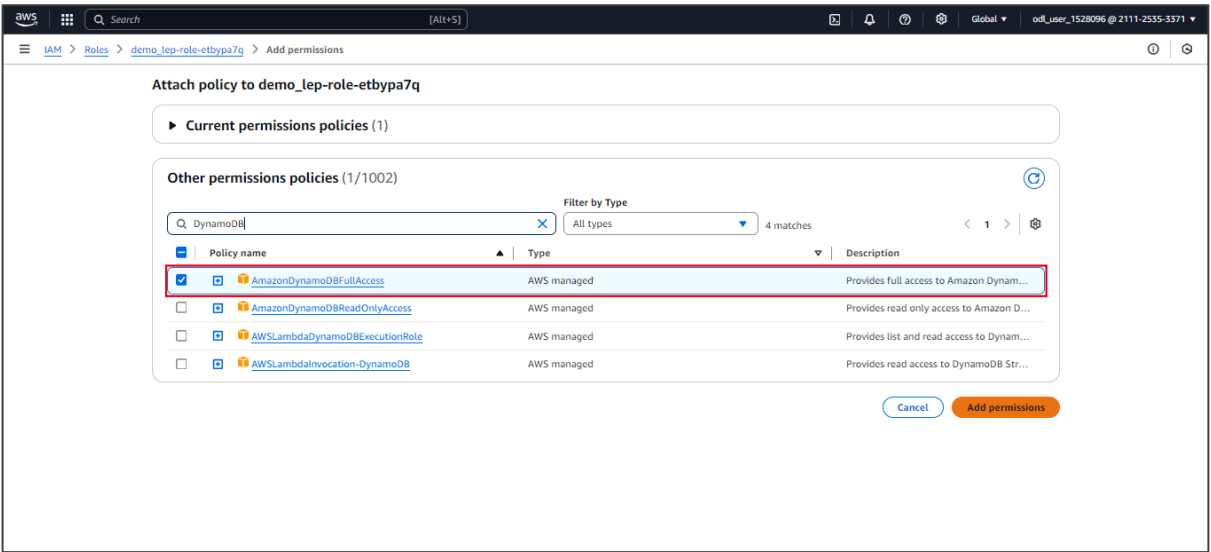
| Policy name                                              | Type             | Attached entities |
|----------------------------------------------------------|------------------|-------------------|
| <code>AWSLambdaBasicExecutionRole-40e7e2c0-39d...</code> | Customer managed | 1                 |

**Permissions boundary** (not set)

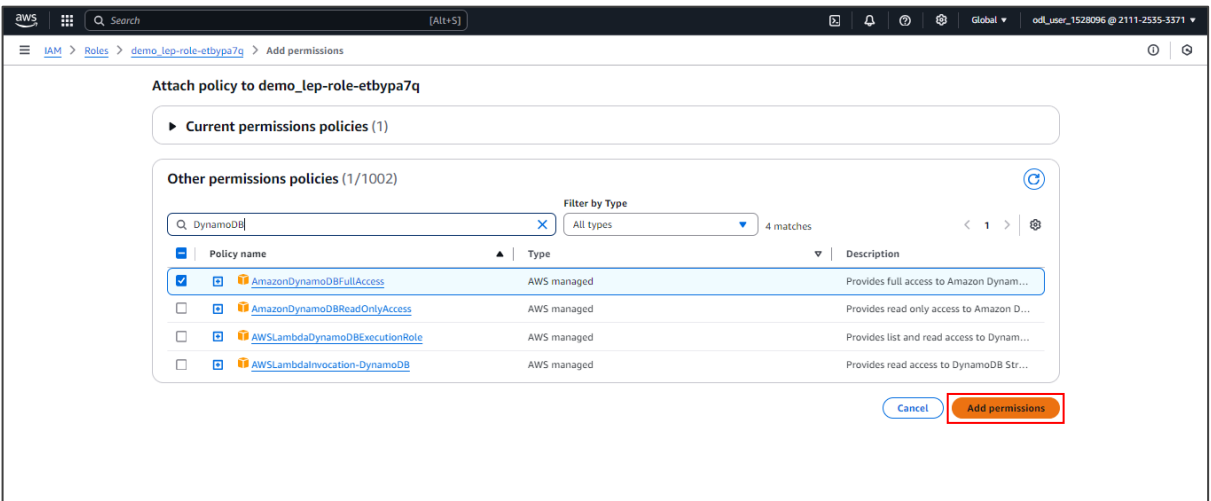
**Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used

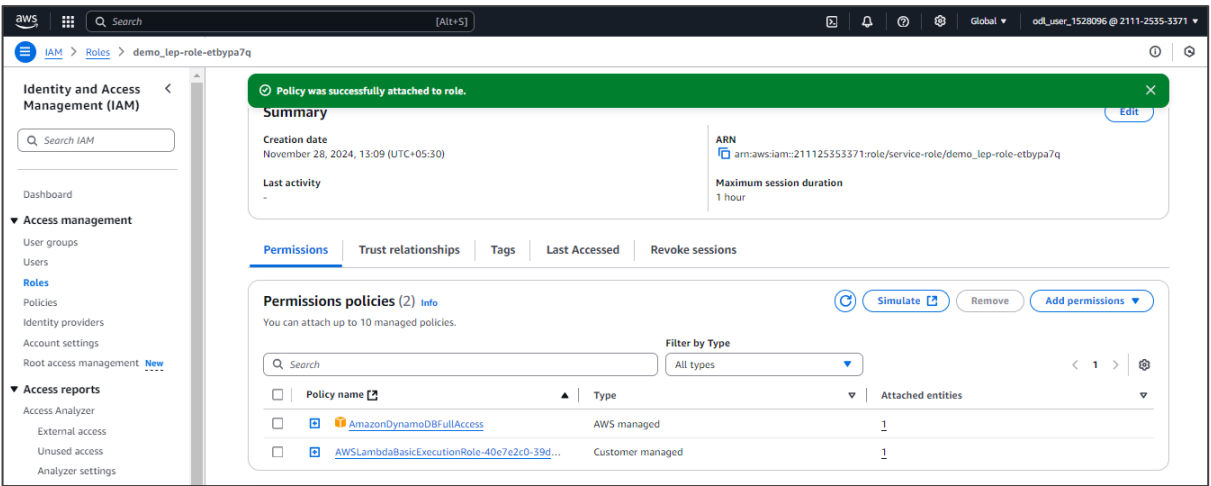
## 2.13 Search **DynamoDB** in the search box and select **AmazonDynamoDBFullAccess**



## 2.14 Click on **Add permission**

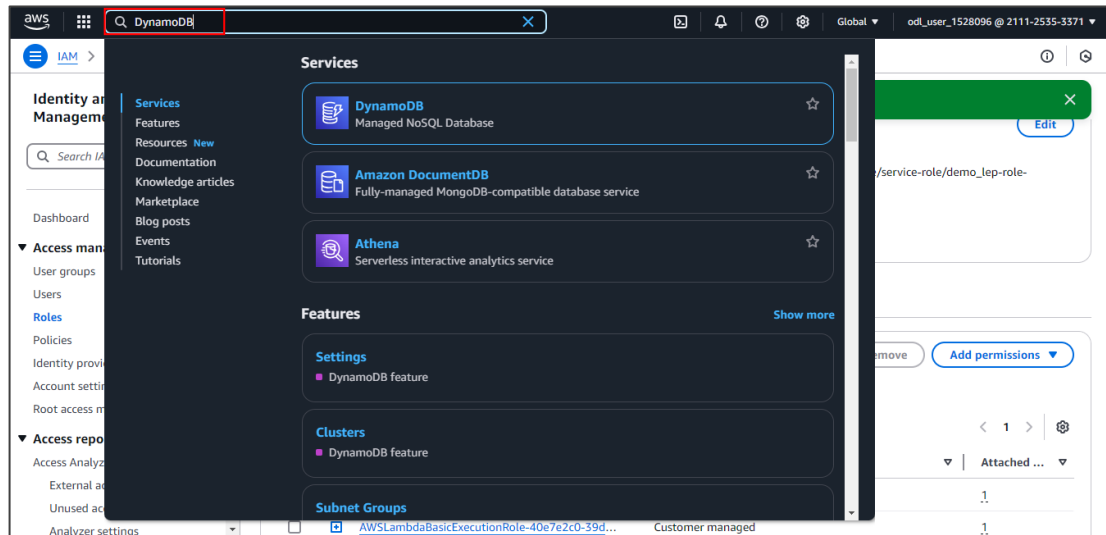


You will see the **Policy was successfully attached to role** message as shown below:

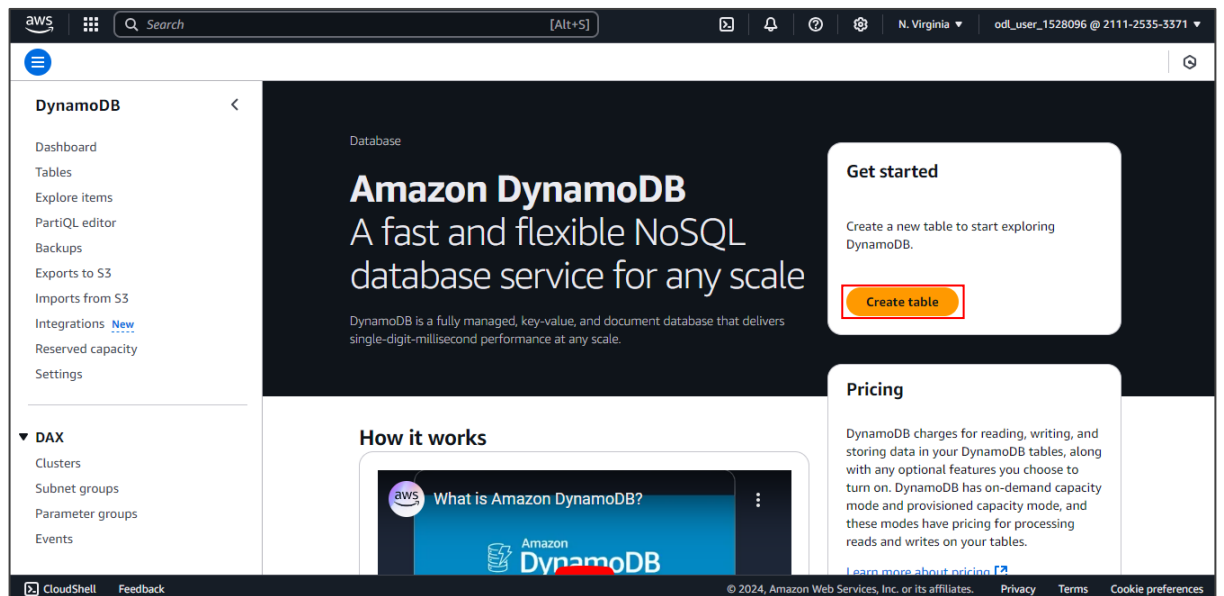


## Step 3: Create a DynamoDB table to store CloudTrail event IDs

### 3.1 Type **DynamoDB** in the search bar and press enter to navigate to DynamoDB Dashboard



### 3.2 Click on **Create table**



### 3.3 Enter Table name as **iamKey\_Storer** and Partition key as **CloudTrail\_key**

**Create table**

**Table details** [Info](#)  
DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

**Table name**  
This will be used to identify your table.  
  
Between 3 and 255 characters, containing only letters, numbers, underscores (\_), hyphens (-), and periods (.).

**Partition key**  
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.  
   
1 to 255 characters and case sensitive.

**Sort key - optional**  
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.  
   
1 to 255 characters and case sensitive.

### 3.4 Scroll down and click on **Create table**

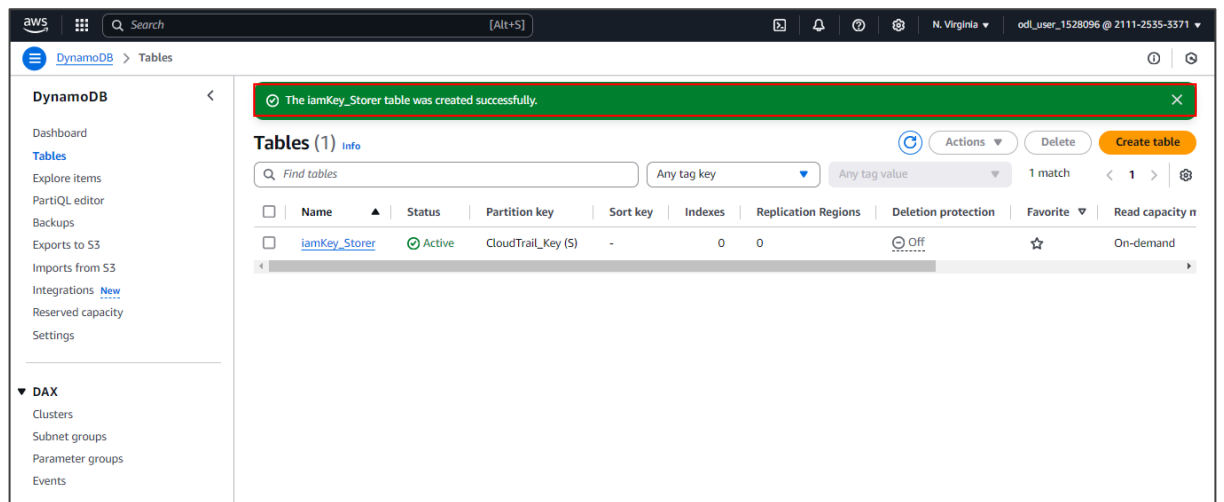
**Configuration options:**

|                              |                          |     |
|------------------------------|--------------------------|-----|
| Maximum write capacity units | -                        | Yes |
| Local secondary indexes      | -                        | No  |
| Global secondary indexes     | -                        | Yes |
| Encryption key management    | Owned by Amazon DynamoDB | Yes |
| Deletion protection          | Off                      | Yes |
| Resource-based policy        | Not active               | Yes |

**Tags**  
Tags are pairs of keys and optional values, that you can assign to AWS resources. You can use tags to control access to your resources or track your AWS spending.  
No tags are associated with the resource.  
[Add new tag](#)  
You can add 50 more tags.

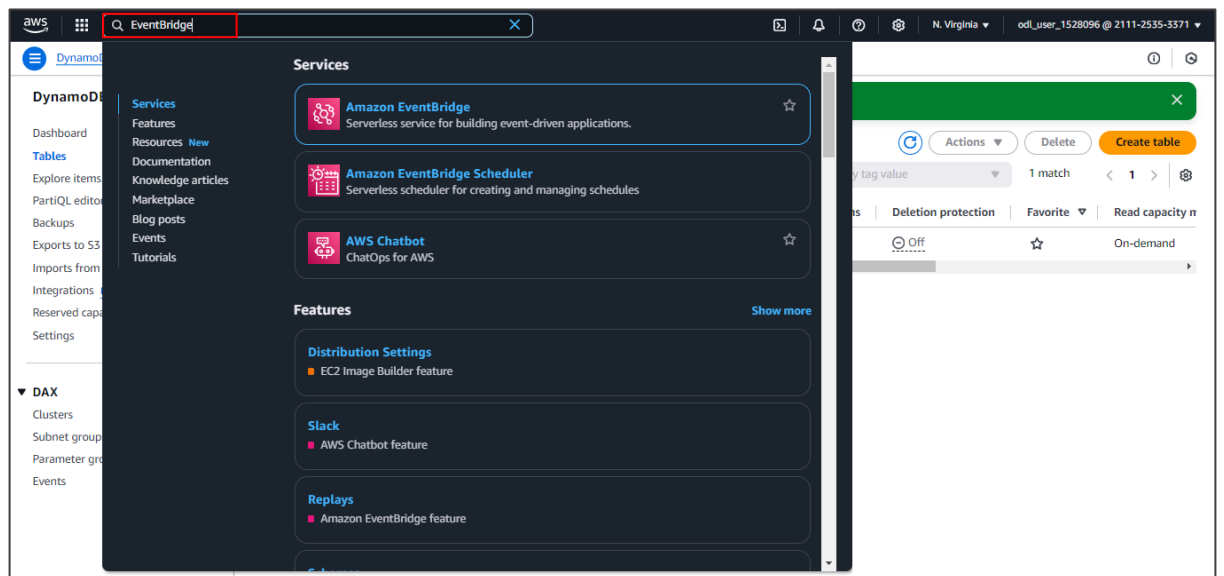
[Cancel](#) [Create table](#)

You will see a **table was created successfully** message box.



## Step 4: Set up an EventBridge rule to trigger the Lambda function

### 4.1 Type **Amazon EventBridge** in the search bar and press enter to navigate to Amazon EventBridge Dashboard



## 4.2 Click on Create rule

The screenshot shows the Amazon EventBridge console. On the left is a navigation menu with sections: Amazon EventBridge, Developer resources, Buses, Pipes, Scheduler, and Integration. The main content area is titled 'How it works' and features a video player for 'Serverless 101: Amazon EventBridge'. To the right of the video is a 'Get started' section with four options: EventBridge Rule (selected), EventBridge Pipes, EventBridge Schedule, and EventBridge Schema registry. The 'Create rule' button is highlighted with a red box. Below this is a 'Pricing' section. At the bottom, there are two boxes: 'Build event-driven architectures' and 'Write less custom code'.

## 4.3 Enter a desired Rule Name and click on Next

The screenshot shows the 'Define rule detail' page in the Amazon EventBridge console. On the left is a step-by-step navigation bar with five steps: Step 1: Define rule detail (selected), Step 2: Build event pattern, Step 3: Select target(s), Step 4 - optional: Configure tags, and Step 5: Review and create. The main content area is titled 'Define rule detail' and contains several fields: 'Name' (with 'rule1' entered and highlighted), 'Description - optional' (with a placeholder 'Enter description'), 'Event bus' (set to 'default'), and 'Rule type' (with 'Rule with an event pattern' selected). At the bottom right, there are 'Cancel' and 'Next' buttons, with the 'Next' button highlighted by a red box.

#### 4.4 Select **AWS services** in the Event source section

The screenshot shows the AWS EventBridge 'Create rule' console. At the top, there are three tabs: 'Use an Amazon EventBridge schema to generate the event pattern.', 'Use a template provided by EventBridge to create an event pattern.', and 'Write an event pattern in JSON.'. The 'Event pattern' section is active. It contains two dropdown menus: 'Event source' (set to 'AWS services') and 'AWS service' (set to 'Select a service provider'). To the right is a large text area for 'Enter the event JSON'. At the bottom right, there are buttons for 'Copy', 'Test pattern', 'Edit pattern', 'Cancel', 'Previous', and 'Next'.

#### 4.5 Select **IAM** in AWS service

This screenshot shows the same AWS EventBridge 'Create rule' console as the previous one, but with the 'AWS service' dropdown menu expanded. The 'IAM' option is highlighted. The 'Event source' dropdown remains set to 'AWS services'. The 'Event type' dropdown is set to 'Select an event type'. The 'Enter the event JSON' text area and the bottom navigation buttons ('Copy', 'Test pattern', 'Edit pattern', 'Cancel', 'Previous', 'Next') are also visible.

#### 4.6 Select **AWS API Call via CloudTrail** in Event type

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Amazon EventBridge > Rules > Create rule

**Event type**  
The type of events as the source of the matching pattern

AWS API Call via CloudTrail

All events that are delivered via CloudTrail have **AWS API Call via CloudTrail** as the value for **detail-type**. Events from API actions that start with the keywords List, Get, or Describe are not processed by EventBridge, with the exception of events from the following STS actions: GetFederationToken and GetSessionToken. Data events (for example, for Amazon S3 object level events, DynamoDB, and AWS Lambda) must have trails configured to receive those events. [Learn more](#).

Copy Test pattern Edit pattern

**Event Type Specification 1**

☐ Any operation

☒ Specific operation(s)

Specific operation(s)

Remove

Add

#### 4.7 Select **Specific operation(s)** in Event Type Specification 1 and enter the Specific operation(s) name **CreateAccessKey**

aws Search [Alt+S] N. Virginia odl\_user\_1528096 @ 2111-2535-3371

Amazon EventBridge > Rules > Create rule

the keywords List, Get, or Describe are not processed by EventBridge, with the exception of events from the following STS actions: GetFederationToken and GetSessionToken. Data events (for example, for Amazon S3 object level events, DynamoDB, and AWS Lambda) must have trails configured to receive those events. [Learn more](#).

Copy Test pattern Edit pattern

**Event Type Specification 1**

☐ Any operation

☒ Specific operation(s)

Specific operation(s)

CreateAccessKey

Remove

Add

Cancel Previous Next

## 4.8 Click on **Next**

the keywords List, Get, or Describe are not processed by EventBridge, with the exception of events from the following STS actions: GetFederationToken and GetSessionToken. Data events (for example, for Amazon S3 object level events, DynamoDB, and AWS Lambda) must have trails configured to receive those events. [Learn more.](#)

**Event Type Specification 1**

☐ Any operation

☒ Specific operation(s)

**Specific operation(s)**

CreateAccessKey

Remove

Add

Cancel Previous **Next**

## 4.9 Select **AWS service** as Target 1 and choose **Lambda function** as the target

**Select target(s)**

**Permissions**  
Note: When using the EventBridge console, EventBridge will automatically configure the proper permissions for the selected targets. If you're using the AWS CLI, SDK, or CloudFormation, you'll need to configure the proper permissions.

**Target 1**

**Target types**  
Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus

☐ EventBridge API destination

☒ **AWS service**

**Select a target** | Info  
Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function

**Function**  
Select or enter a Lambda function ARN

Configure version/alias

#### 4.10 Select the Lambda function created in **Step 2**

aws

Search

[Alt+S]

N. Virginia

odl\_user\_1528096 @ 2111-2535-3371

Amazon EventBridge > Rules > Create rule

Step 2  
Select target(s)

Step 4 - optional  
Configure tags

Step 5  
Review and create

the AWS CLI, SDK, or CloudFormation, you'll need to configure the proper permissions.

Target 1

Target types

Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus

☐ EventBridge API destination

☒ AWS service

Select a target | Info

Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function

Function

demo\_lep

Q |

Enter the Lambda function ARN

Select Lambda function

demo\_lep

End of results

Next

#### 4.11 Click on **Next**

[Alt+S]

N. Virginia

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Create rule

Select target(s)

Step 4 - optional  
Configure tags

Step 5  
Review and create

the AWS CLI, SDK, or CloudFormation; you'll need to configure the proper permissions.

### Target 1

**Target types**  
Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus  
☐ EventBridge API destination  
☒ AWS service

**Select a target** | Info  
Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function ▼

**Function**

demo\_lep ▼ ↻

► Configure version/alias

► Additional settings

Add another target

Cancel

Skip to Review and create

Previous

Next

## 4.12 Click on **Remove**

The screenshot shows the AWS EventBridge console during the 'Create rule' process. The left sidebar lists five steps: 'Define rule detail', 'Build event pattern', 'Select target(s)', 'Configure tags' (which is the active step, indicated by a blue circle), and 'Review and create'. The main content area is titled 'Configure tags - optional' with an 'Info' link. Below the title, there is a description of tags and two input fields: 'Key' and 'Value - optional'. The 'Remove' button is highlighted with a red box. At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Next'.

aws [Search] [Alt+S] N. Virginia odLuser\_1528096 @ 2111-2535-3371

Amazon EventBridge > Rules > Create rule

Step 1 Define rule detail  
Step 2 Build event pattern  
Step 3 Select target(s)  
Step 4 - optional **Configure tags**  
Step 5 Review and create

### Configure tags - optional [Info](#)

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key**  
Enter key

**Value - optional**  
Enter value

**Remove**

**Add new tag**  
You can add 49 more tags

Cancel Previous **Next**

## 4.13 Click on **Next**

The screenshot shows the same AWS EventBridge console as the previous one, but now the 'Next' button is highlighted with a red box. The 'Add new tag' button is also visible. The rest of the interface remains the same.

aws [Search] [Alt+S] N. Virginia odLuser\_1528096 @ 2111-2535-3371

Amazon EventBridge > Rules > Create rule

Step 1 Define rule detail  
Step 2 Build event pattern  
Step 3 Select target(s)  
Step 4 - optional **Configure tags**  
Step 5 Review and create

### Configure tags - optional [Info](#)

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

**Add new tag**  
You can add 50 more tags

Cancel Previous **Next**

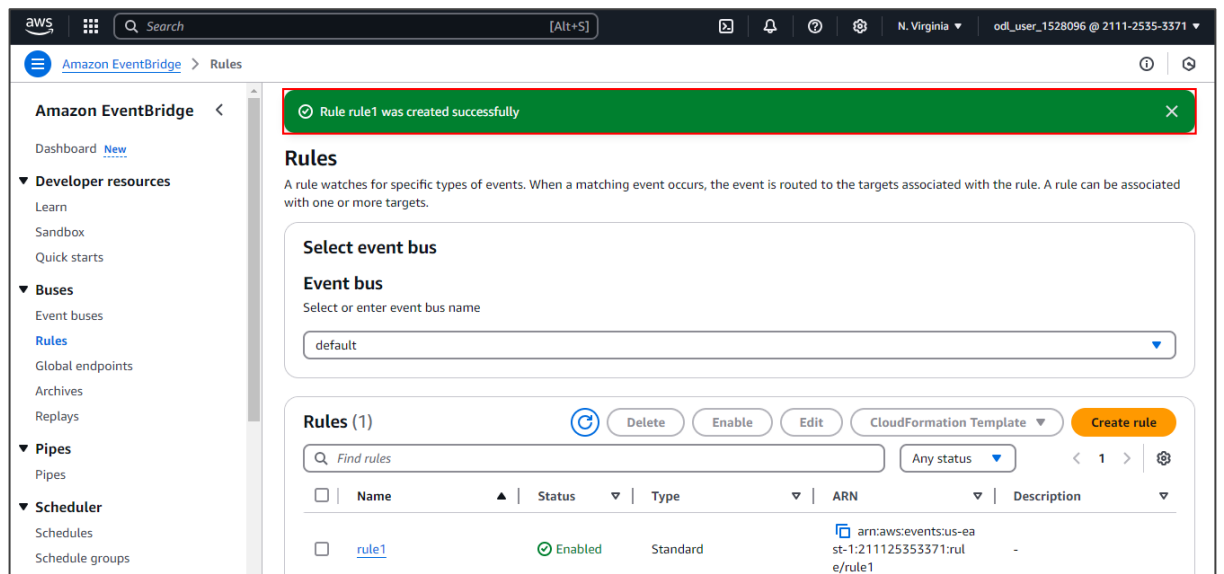
#### 4.14 Scroll down in the **Review and create** page and click on **Create rule**

The screenshot shows the AWS EventBridge console at the 'Review and create' stage for creating a new rule. The left sidebar lists five steps: Step 1 (Define rule detail), Step 2 (Build event pattern), Step 3 (Select target(s)), Step 4 (optional, Configure tags), and Step 5 (Review and create). Step 1 is currently active. The main content area is titled 'Review and create' and 'Step 1: Define rule detail'. It contains a 'Define rule detail' section with three fields: 'Rule name' (rule1), 'Status' (Enabled with a green checkmark), and 'Event bus' (default). Below this is the 'Step 2: Build event pattern' section, which shows an 'Event pattern' field with a JSON configuration: 

```
1 {
2   "source": ["aws.iam"],
3   "detail-type": ["AWS API Call via CloudTrail"],
4   "detail": {
5     "eventSource": ["iam.amazonaws.com"],
6     "eventName": ["CreateAccessKey"]
7   }
8 }
```

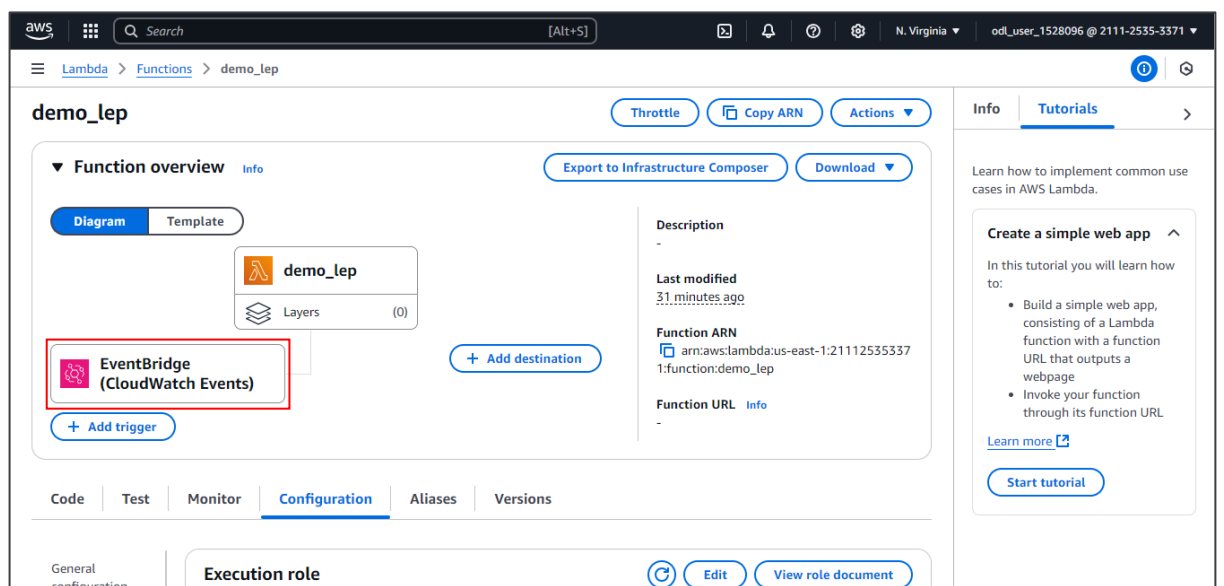
The screenshot shows the AWS EventBridge console at the 'Review and create' stage, specifically Step 4: Configure tag(s). The left sidebar shows the same five steps as the previous screenshot, with Step 4 now active. The main content area is titled 'Step 4: Configure tag(s)'. It features a 'Targets' section with a table showing one target: 'demo\_lep' (Lambda function) with ARN 'arn:aws:lambda:us-east-1:211125353371:function:demo\_lep' and input 'Matched event'. Below the table, there are fields for 'Input to target:' (Matched event), 'Additional parameters:' (empty), and 'Dead-letter queue (DLQ):' (empty). At the bottom, there is a 'Tags (0)' section with a description: 'A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.' Below this is a table with columns 'Key' and 'Value', and a message 'No tags associated with this resource.' At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Create rule' (highlighted in orange).

You will see the following message box of successful rule creation.

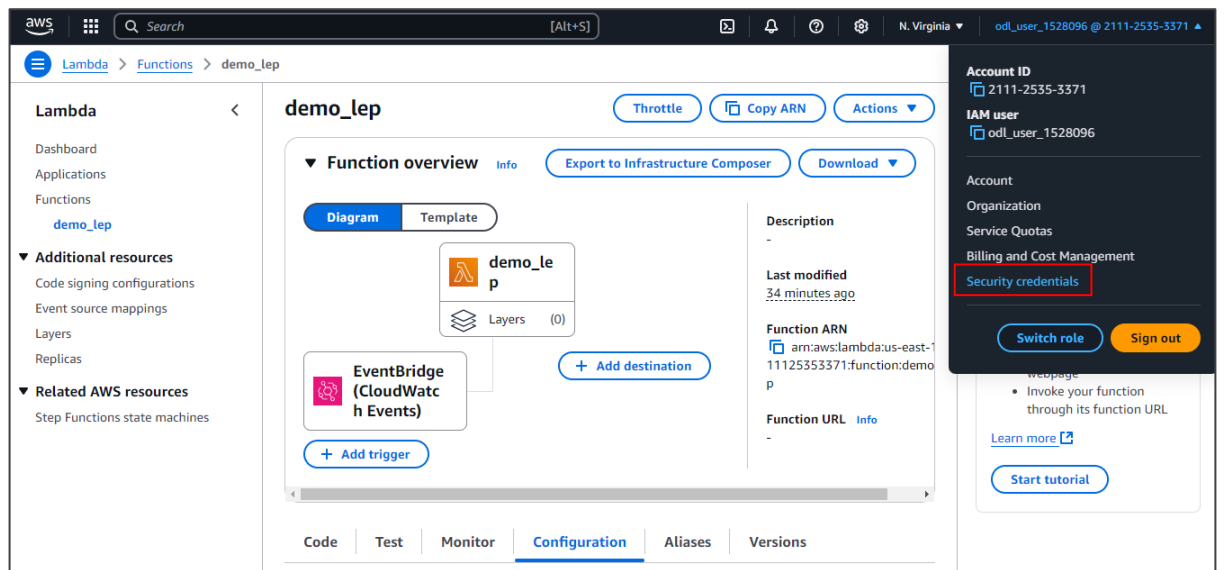


## Step 5: Generate access and secret keys in IAM for testing

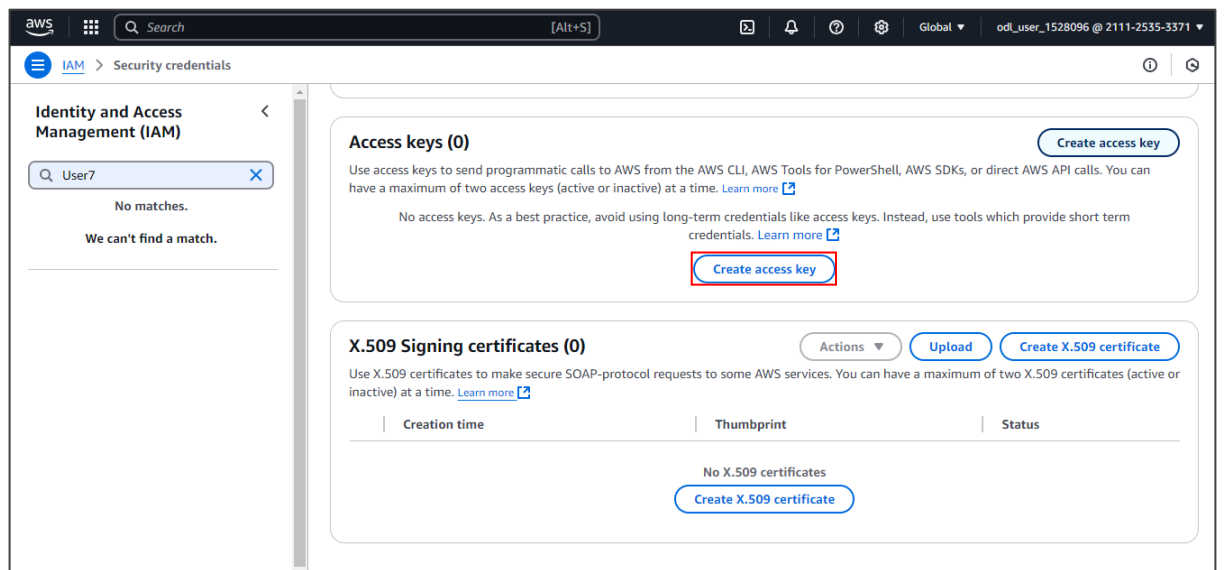
### 5.1 Navigate to the Lambda function page to view the trigger addition



## 5.2 Click on the Account settings tab and click on **Security credentials**



## 5.3 Scroll down and click on **Create access key**



## 5.4 Select Command Line Interface (CLI)

aws | Search [Alt+S] | Global | odouser\_1528096 @ 2111-2535-3371

IAM > Security credentials > Create access key

Step 1: Access key best practices & alternatives (selected)  
Step 2 - optional: Set description tag  
Step 3: Retrieve access keys

### Access key best practices & alternatives [Info](#)

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

**Use case**

- ☒ **Command Line Interface (CLI)**  
You plan to use this access key to enable the AWS CLI to access your AWS account.
- ☐ **Local code**  
You plan to use this access key to enable application code in a local development environment to access your AWS account.
- ☐ **Application running on an AWS compute service**  
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.
- ☐ **Third-party service**  
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.
- ☐ **Application running outside AWS**  
You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.

## 5.5 Select the checkbox and click on Next

aws | Search [Alt+S] | Global | odouser\_1528096 @ 2111-2535-3371

IAM > Security credentials > Create access key

**Alternatives recommended**

- Use [AWS CloudShell](#), a browser-based CLI, to run commands. [Learn more](#)
- Use the [AWS CLI V2](#) and enable authentication through a user in IAM Identity Center. [Learn more](#)

**Confirmation**

☒ I understand the above recommendation and want to proceed to create an access key.

[Cancel](#) [Next](#)

## 5.6 Click on **Create access Key**

The screenshot shows the AWS IAM console 'Create access key' page. On the left, a progress bar indicates three steps: Step 1 (Access key best practices & alternatives), Step 2 (Set description tag - optional), and Step 3 (Retrieve access keys). Step 2 is currently selected. The main content area is titled 'Set description tag - optional' with an 'info' icon. Below the title, it states: 'The description for this access key will be attached to this user as a tag and shown alongside the access key.' There is a text input field for the 'Description tag value' with a placeholder text: 'Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.' Below the input field, a note says: 'Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: \_ . : / \* + - @'. At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Create access key' (which is highlighted with a red rectangle).

You will see the **Access Key created** message.

The screenshot shows the AWS IAM console 'Create access key' page, Step 3: Retrieve access keys. A green banner at the top states: 'Access key created. This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.' The progress bar on the left shows Step 3 as the current step. The main content area is titled 'Retrieve access keys' with an 'info' icon. Below the title, it states: 'Access key. If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.' There are two fields: 'Access key' with the value 'AKIATCKANHONXLTJXJFL' and 'Secret access key' with a masked value '\*\*\*\*\*' and a 'Show' button. Below these fields, there is a section titled 'Access key best practices' with a list of bullet points: 'Never store your access key in plain text, in a code repository, or in code.', 'Disable or delete access key when no longer needed.', 'Enable least-privilege permissions.', and 'Rotate access keys regularly.' A link says: 'For more details about managing access keys, see the best practices for managing AWS access keys.' At the bottom right, there are two buttons: 'Download .csv file' and 'Done'.

## Step 6: Monitor Lambda execution and CloudTrail logs in CloudWatch

### 6.1 Navigate to the DynamoDB table created in Step 3

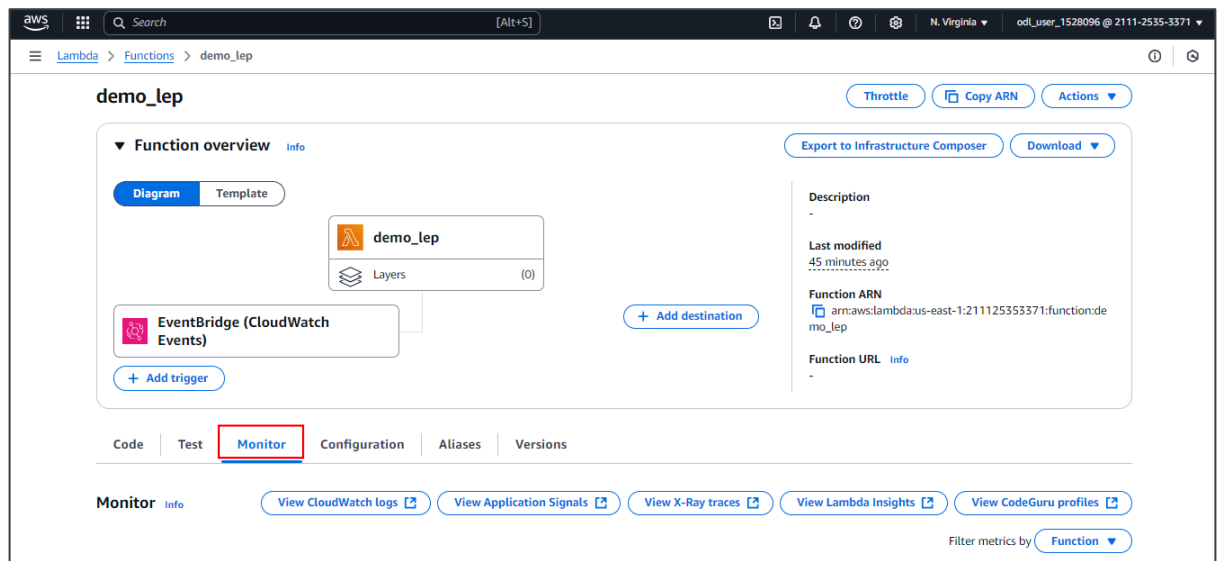
The screenshot shows the AWS Management Console interface for a DynamoDB table named 'iamKey\_Storer'. The left sidebar contains the 'DynamoDB' menu with options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The 'Tables' section is active, showing a list of tables with 'iamKey\_Storer' selected. The main content area displays the 'Overview' tab for 'iamKey\_Storer'. It includes a warning about PITR, general information (Partition key: CloudTrail\_Key, Sort key: -, Capacity mode: On-demand, Table status: Active), and an 'Items summary' section.

### 6.2 Click on **Explore items**

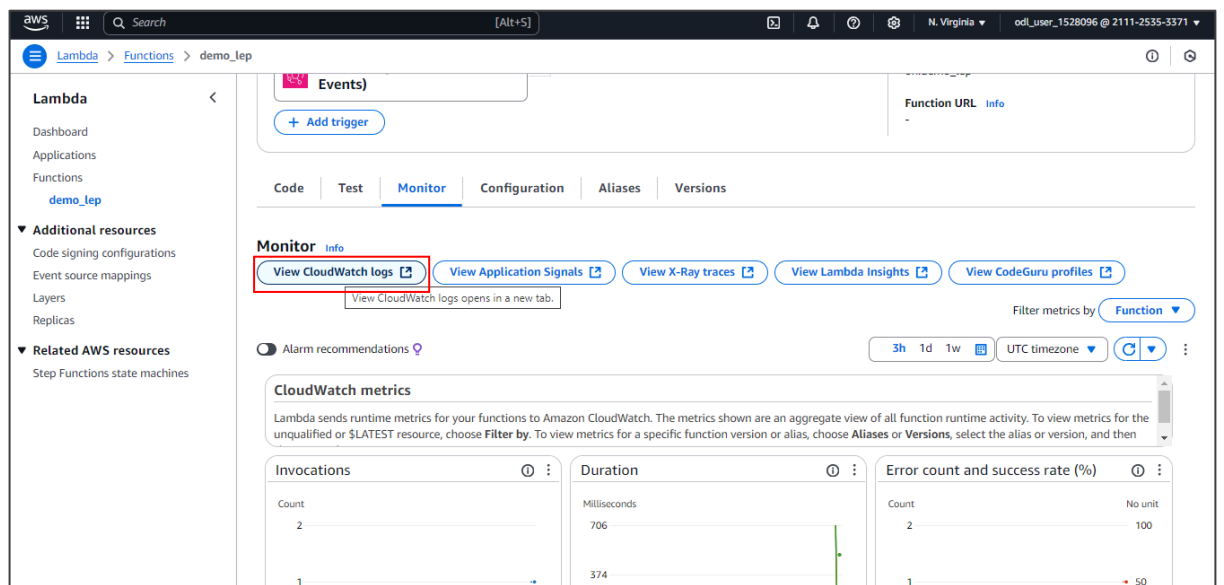
The screenshot shows the 'Explore items' page for the 'iamKey\_Storer' table. The left sidebar is the same as in the previous screenshot, but the 'Explore items' option is highlighted with a red box. The main content area shows the 'Scan or query items' section with a 'Scan' button selected. Below this, there are fields for 'Select a table or index' (Table - iamKey\_Storer) and 'Select attribute projection' (All attributes). A 'Run' button is visible. A green status bar indicates 'Completed. Read capacity units consumed: 2'. The 'Items returned (0)' section shows no items.

**Note:** Verify the entries in the table after logs are monitored

### 6.3 Navigate to the Lambda function page and click on **Monitor** tab



### 6.4 Click on **View CloudWatch logs**



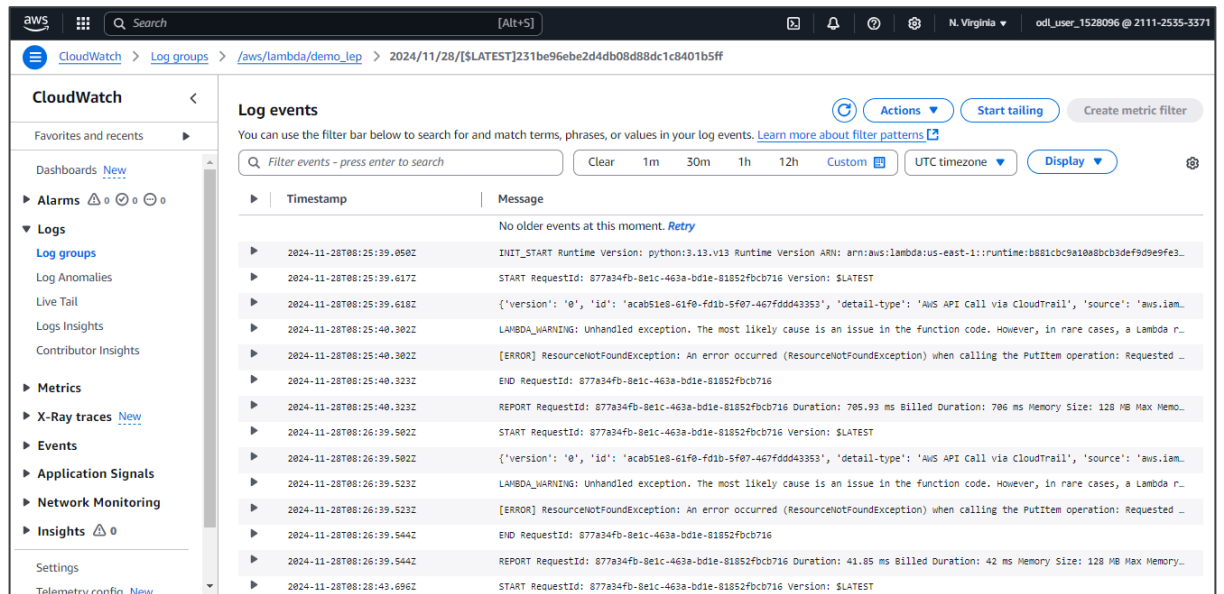
## 6.5 Select the **Log streams** tab

The screenshot shows the AWS CloudWatch console interface. The left sidebar contains navigation options: CloudWatch, Favorites and recents, Dashboards, Alarms, Logs, Metrics, X-Ray traces, Events, Application Signals, Network Monitoring, and Insights. The main content area is titled '/aws/lambda/demo\_lep'. Below the title, there are buttons for 'Actions', 'View in Logs Insights', 'Start tailing', and 'Search log group'. The 'Log group details' section displays various attributes: Log class (Standard), ARN (arn:aws:logs:us-east-1:211125353371:log-group:/aws/lambda/demo\_lep:\*), Creation time (7 minutes ago), Retention (Never expire), Stored bytes (-), Metric filters (0), Subscription filters (0), Contributor Insights rules (-), KMS key ID (-), Anomaly detection (Configure), Data protection (-), Sensitive data count (-), Field indexes (Configure), and Transformer (Configure). Below this, a horizontal tab bar includes 'Log streams' (highlighted with a red box), 'Tags', 'Anomaly detection', 'Metric filters', 'Subscription filters', 'Contributor Insights', 'Data protection', and 'File'. The 'Log streams (1)' section shows a search bar with the placeholder 'Filter log streams or try prefix search' and buttons for 'Delete', 'Create log stream', and 'Search all log streams'. At the bottom, there are checkboxes for 'Exact match' and 'Show expired', along with an 'Info' link and pagination controls showing '1' of 1 items.

## 6.6 Click on the log stream created

This screenshot shows the same AWS CloudWatch console interface as the previous one, but with the 'Log streams' tab selected. The 'Log streams (1)' section now displays a list of log streams. The first and only entry is a log stream with the name '2024/11/28/[\$LATEST]231be96be2d4db08d88dc1c8401b5ff', which is highlighted with a red box. To the right of the log stream name, the 'Last event time' is shown as '2024-11-28 08:25:40 (UTC)'. The rest of the interface, including the sidebar and the 'Log group details' section, remains the same.

## 6.7 Verify the Log events



The screenshot shows the AWS CloudWatch console interface. The left sidebar contains navigation links for CloudWatch, Favorites and recents, Dashboards, Alarms, Logs, Metrics, X-Ray traces, Events, Application Signals, Network Monitoring, and Insights. The main content area is titled 'Log events' and shows a list of log events for the Lambda function 'aws/lambda/demo\_lep'. The events are filtered by the log group '2024/11/28/[LATEST]231be96be2d4db08d88dc1c8401b5ff'. The events are displayed in a table with columns for Timestamp and Message. The events include INIT\_START, START, LAMBDA\_WARNING, [ERROR], and END messages, along with their respective RequestId and Version information.

| Timestamp                | Message                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 2024-11-28T08:25:39.050Z | INIT_START Runtime version: python:3.13.v13 Runtime Version ARN: arn:aws:lambda:us-east-1:runtime:b881c9a10a8bc3def9d9e9fe3...     |
| 2024-11-28T08:25:39.617Z | START RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716 Version: \$LATEST                                                             |
| 2024-11-28T08:25:39.618Z | {'version': '0', 'id': 'acab51e8-61f0-fd1b-5f07-467fddd43353', 'detail-type': 'AWS API Call via CloudTrail', 'source': 'aws.iam... |
| 2024-11-28T08:25:40.302Z | LAMBDA_WARNING: Unhandled exception. The most likely cause is an issue in the function code. However, in rare cases, a Lambda r... |
| 2024-11-28T08:25:40.302Z | [ERROR] ResourceNotFoundException: An error occurred (ResourceNotFoundException) when calling the PutItem operation: Requested ... |
| 2024-11-28T08:25:40.323Z | END RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716                                                                                 |
| 2024-11-28T08:25:40.323Z | REPORT RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716 Duration: 705.93 ms Billed Duration: 706 ms Memory Size: 128 MB Max Memo...  |
| 2024-11-28T08:26:39.580Z | START RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716 Version: \$LATEST                                                             |
| 2024-11-28T08:26:39.580Z | {'version': '0', 'id': 'acab51e8-61f0-fd1b-5f07-467fddd43353', 'detail-type': 'AWS API Call via CloudTrail', 'source': 'aws.iam... |
| 2024-11-28T08:26:39.523Z | LAMBDA_WARNING: Unhandled exception. The most likely cause is an issue in the function code. However, in rare cases, a Lambda r... |
| 2024-11-28T08:26:39.523Z | [ERROR] ResourceNotFoundException: An error occurred (ResourceNotFoundException) when calling the PutItem operation: Requested ... |
| 2024-11-28T08:26:39.544Z | END RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716                                                                                 |
| 2024-11-28T08:26:39.544Z | REPORT RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716 Duration: 41.85 ms Billed Duration: 42 ms Memory Size: 128 MB Max Memory...  |
| 2024-11-28T08:28:43.696Z | START RequestId: 877a34fb-8e1c-463a-bd1e-81852fbc716 Version: \$LATEST                                                             |

By following the above steps, you have successfully automated the monitoring of AWS IAM access key creation. This includes creating a CloudTrail to log IAM activities, deploying a Lambda function to process access key creation events, setting up a DynamoDB table to store event data, and configuring an EventBridge rule to trigger the Lambda function.